

Filtrotecnica FRM 1140x2500 - calculated airflow and pressure drop

Purchased automatic roll filter - project design data for the process-air system.

FRM 1140 × 2500 Horizontal installation G3 media

Confirmed purchased configuration

Parameter	Confirmed value
Model / article	FRMZ002511 - Automatic rotary filter 1140 × 2500 FRM
Installation	Horizontal, direct crossing, with housing and double flange
Design airflow	14,600 m³/h
Air crossing speed	2 m/s
Process condition	46 °C / 91.5% RH
Filter media	Fiberglass, reinforced scrim
Efficiency	G3 class (EN 779)
Suggested maximum pressure drop	150 Pa
Roll-drive motor	0.06 kW, 400 V / 50 Hz, IP55
Controls	Differential-pressure switch + end-roll microswitch + IP65 control box

Pressure-drop interpretation

The delivery document does not specify the factory setpoint at which the roll advances. It only confirms a differential-pressure switch and a suggested maximum pressure drop of 150 Pa.

Clean-filter Δp Not specified	Suggested maximum Δp 150 Pa	Design airflow 14,600 m³/h
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Crossing speed
2.0 m/s

For fan sizing, the conservative value is therefore:

$\Delta p_{\text{filter, design}} = 150 \text{ Pa}$

Recommended commissioning logic

Condition	Engineering setting	Status
Normal operation	Below roll-advance threshold	Project logic
Start advancing filter media	120-130 Pa	Recommended commissioning range - not manufacturer specified
High pressure-drop alarm	150 Pa	Matches supplier's suggested maximum

Illustration: 130 Pa is 86.7% of the 150 Pa maximum design pressure drop.

Flow check above the quoted design point

If the process-air system runs at **15,700 m³/h**, and if face velocity scales proportionally with flow, the corresponding crossing speed would be approximately **2.15 m/s**.

This does **not** establish that 15,700 m³/h is a supplier-approved operating point. The purchased selection is documented at 14,600 m³/h and 2 m/s.

Use in the system pressure budget

For the worst-case fan calculation, reserve the full **150 Pa** for this filter. Add this to the pressure losses of the HOLTOP recuperator, ducts, bends, transitions and outlet components. Do not use an assumed clean-filter value for final fan selection unless it is measured or supplied by Filtrotecnica.

Source: Filtrotecnica delivery document no. 651, order #P00441, job 2026/6839. All recommendations beyond the quoted 150 Pa maximum are project engineering assumptions and should be confirmed during commissioning.